

AI508

Computer

Vision

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Computer vision: definition

GENERAL DEFINITION:

The study and practice of learning and computing with visual data to infer, organize, and synthesize information about the visual world.

SPECIFIC DEFINITION:

The study and practice of developing and applying deep learning models for perceiving and generating visual and multi-modal data.

Here, visual data refers to images, videos, and other 2D and 3D signals.

Computer vision is a multi-disciplinary field building on, e.g., computer science, mathematics, psychology, neurobiology, electrical engineering, and statistics.



Computer vision: fields

Image processing (denoising, deblurring, super-resolution, color models)

Low-level vision (edge/line/corner/feature detection)

Geometry (stereo vision, depth/pose estimation, multi-view geometries)

Perception (object detection, classification, segmentation)

Synthesis (image/video generation, in/outpainting, style transfer)

Multimodal (contrastive language-image pre-training, vision language models)



Computer vision: tasks

Image classification

Object detection

Segmentation

Image generation

Inpainting

Image-text retrieval

Visual question answering

Video generation

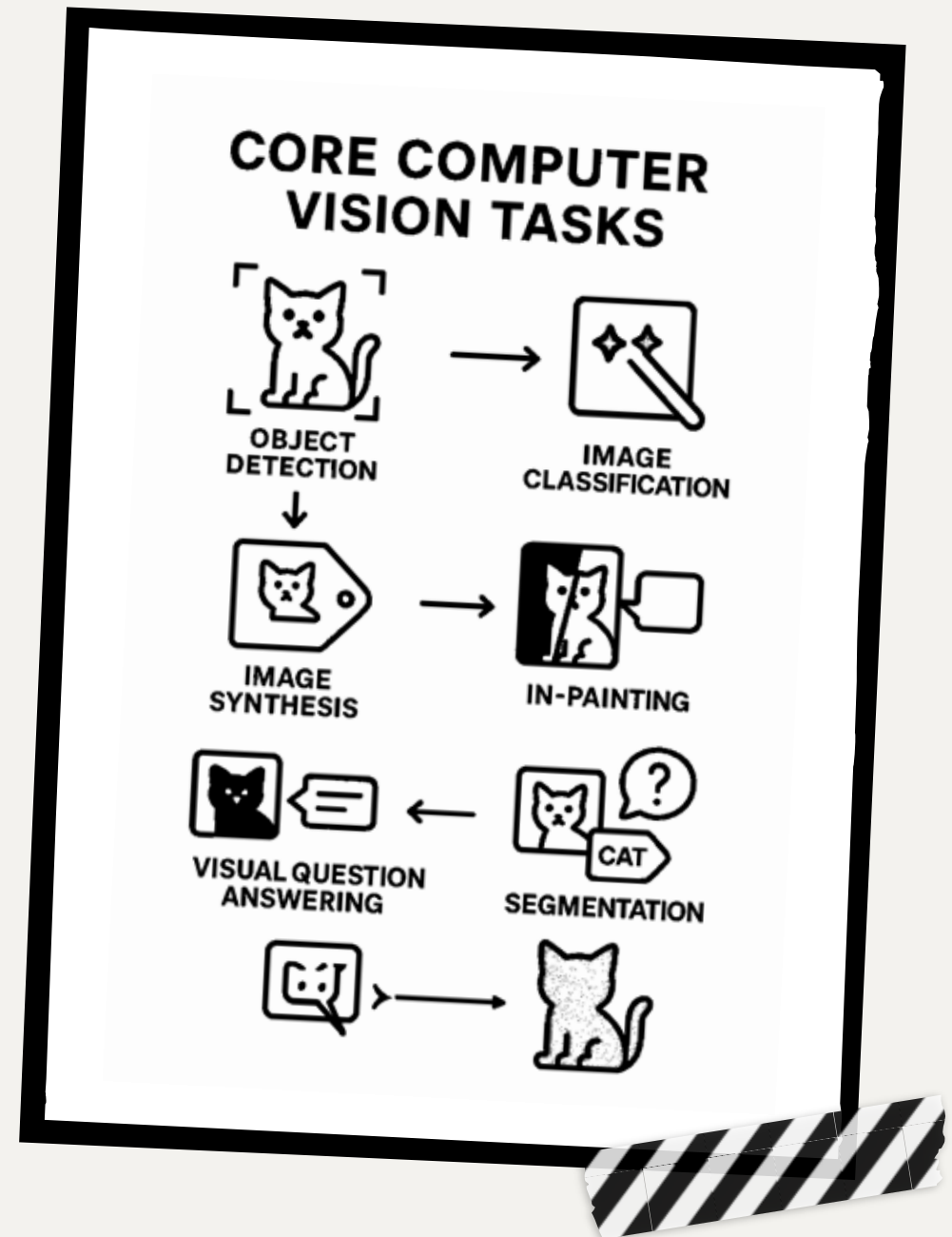


Image classification

GOAL:

Assign a (typically single) label to an image.

METRICS:

Accuracy (top-k), confusion matrix.

DATASETS:

CIFAR-10, CIFAR-100, ImageNet.

ARCHITECTURES:

Convolutional Neural Networks, Vision Transformer.

? DOG ?



? MUFFIN ?



Object detection

GOAL :

Localize objects on an image.

METRICS:

IoU, mean Average Precision.

DATASETS:

COCO, VOC.

ARCHITECTURES:

R-CNN, YOLO.



Segmentation

GOAL :

Predict one label per pixel.

METRICS:

mIoU, AP, PQ.

DATASETS:

COCO-Seg, SA-1B, SA-V.

ARCHITECTURES:

UNet, SAM, SOLOv2.



Image generation

GOAL:

Synthesize realistic images.

METRICS:

FID, KID, CLIPScore; human preference.

DATASETS:

LAION-5B, COCO, ImageNet.

ARCHITECTURES:

Diffusion, VAE, GAN.



Inpainting

GOAL:

Fill masked regions coherently.

METRICS:

PSNR, SSIM, LPIPS.

DATASETS:

Places2, CelebA-HQ.

ARCHITECTURES:

Diffusion, Partial/Gated/Dilated CNNs.



Image-text retrieval

GOAL :

Embed images and text jointly.

METRICS:

Recall, mean Average Precision.

DATASETS:

CLIP, LAION.

ARCHITECTURES:

CLIP, BLIP, ViLBERT.

	Dataset Examples	ImageNet	Zero-Shot	Δ Score
		ResNet101	CLIP	
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%



Visual question answering

GOAL :

Answer questions regarding an image.

METRICS:

Accuracy, LLM-as-a-judge.

DATASETS:

VQAv2, GQA, TextVQA, OK-VQA.

ARCHITECTURES:

VLM (CLIP/BLIP + LLM).



What does this image show?

You're standing in Plaza del Ayuntamiento, Valencia, in front of the Ayuntamiento (City Hall). Blue hour shot, strong mixed lighting, wide-angle phone look, heavy highlight clipping on the façade.



Video generation

GOAL :

Generate temporally coherent videos.

METRICS:

FVD, KID, CLIPScore, T2VScore.

DATASETS:

WebVid-10M, UCF101, Kinetics.

ARCHITECTURES:

LVD, VideoGPT, DiT.



The end ...

is just the beginning!

LECTURE 1: Introduction

LECTURE 2: Vision Transformer

LECTURE 3: You Only Look Once

LECTURE 4: Diffusion

LECTURE 5: Segment Anything

LECTURE 6: Vision Language Models

